

Md Raihan Khan

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EDUCATION

Khulna University of Engineering and Technology

Khulna, Bangladesh

Bachelor of Science in Electrical and Electronic Engineering

January 2019 – March 2024

Cumulative Grade Point Average: 3.71 / 4.00, ranked 24 of 149 (top 16%)

Thesis: Assessing Mental Stress Using Deep Learning Methods from EEG Signals

Supervisor: Dr. Mohiuddin Ahmad, Professor of Electrical and Electronic Engineering

RESEARCH INTERESTS

- One-step and few-step generative modeling, flow matching, and Schrödinger bridges
- Uncertainty quantification for reliable generative models
- Optimal transport and probabilistic methods for efficient generation
- Geometry-aware deep learning and structured representations

RESEARCH EXPERIENCE

Research Collaborator

September 2026 – Present

Vision and Learning Lab, Clemson University

Supervisor: Dr. Siyu Huang; theoretical collaborator: Dr. Minhyuk Sung (KAIST, remote)

- Identified a gap in one-step generative models and proposed PT-Flow, a Schrödinger-bridge formulation with single-evaluation generation.
- Formulated the prox-tilted importance estimator and refined its theoretical analysis through iterative remote collaboration with Dr. Sung.
- Conducted ImageNet-1K experiments on Clemson's Palmetto cluster, optimizing code and experimental settings. PT-Flow achieved an FID of 1.54 on ImageNet 256×256 with one function evaluation; the resulting paper was submitted to ICLR 2027 and is under review.

Research Collaborator

November 2024 – July 2026

Department of Electrical and Electronic Engineering, Khulna University of Engineering and Technology

Advisor: Dr. Mohiuddin Ahmad

- Designed AutoLumNet, a bi-branch exposure-aware network performing single-shot correction of both under- and over-exposed images; reformulated the correction step as monotone optimal transport, attaining an average MSEC score of 23.75.
- Developed DAFNet, fusing a local 2D-CNN branch and a global graph-attention transformer with covariance pooling, reaching state-of-the-art accuracy on multiple EEG emotion benchmarks.

Undergraduate Thesis Researcher

January 2023 – March 2024

Department of Electrical and Electronic Engineering, Khulna University of Engineering and Technology

Supervisor: Dr. Mohiuddin Ahmad

- Showed phase-locking and cross-correlation features favor larger kernels, while wavelet energy and differential entropy favor smaller ones, in 3D CNNs for EEG emotion recognition.
- Built the complete preprocessing, class-balancing, and evaluation pipeline on constrained hardware without GPU access.

PUBLICATIONS

Manuscripts Under Review

Md Raihan Khan, Run Wang, Siyu Huang, and Minhyuk Sung, “PT-Flow: One-Step Schrödinger-Bridge Generation by Proximal Tilting,” *under review at the Fifteenth International Conference on Learning Representations (ICLR 2027)*.

Airin Akter Tania, **Md Raihan Khan**, and Mohiuddin Ahmad, “AutoLumNet: Monotone optimal transport for single-shot exposure correction,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *under review*, 2026.

Md Raihan Khan and Airin Akter Tania, “Equivariant geodesic networks: End-to-end classification on the SPD manifold,” *The 30th International Conference on Artificial Intelligence and Statistics, under review*, 2027.

Refereed Journal Articles

Md Raihan Khan, Airin Akter Tania, Tanjum Arifen Bushra, Jahanara Pritha, and Mohiuddin Ahmad, “DAFNet: A dual-path attention fusion network for EEG emotion recognition via CNN and graph-based global modeling,” *Array (Elsevier)*, 2025.

Md Raihan Khan, Airin Akter Tania, and Mohiuddin Ahmad, “A comparative study of time–frequency features based spatio-temporal analysis with varying multiscale kernels for emotion recognition from EEG,” *Biomedical Signal Processing and Control (Elsevier)*, 2025.

Refereed Conference Proceedings

Md Rifat Ur Rahman, **Md Raihan Khan**, Md Sakib Hossain Shovon, Pietro Liò, and Mohammad Ali Moni, “Orthogonal polynomial approximation for matrix log normalization in global covariance pooling,” *37th British Machine Vision Conference (BMVC)*, *accepted*, 2026.

Airin Akter Tania and **Md Raihan Khan**, “Difficulty-calibrated interpolation paths for conditional flow matching,” *Third IEEE International Conference on Computing, Applications and Systems*, *accepted*, 2026.

Airin Akter Tania, **Md Raihan Khan**, and Mohiuddin Ahmad, “Custom dataset-driven unsupervised low-light image enhancement using 2D CNN,” *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking*, 2025.

Md Raihan Khan and Mohiuddin Ahmad, “Mental stress detection from EEG signals using comparative analysis of random forest and recurrent neural network,” *IEEE International Conference on Advances in Computing, Communication, Electrical, and Smart Systems*, 2024.

Md Raihan Khan, Abdullah Al Ahad, Airin Akter Tania, Tithi Das, and Bibhakar Das, “Smart classroom automation: A fusion of AI with voice, gesture, and face recognition attendance system,” *IEEE International Conference on Advances in Computing, Communication, Electrical, and Smart Systems*, 2024.

SOFTWARE AND OPEN-SOURCE CONTRIBUTIONS

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| egnlb — Equivariant Geodesic Networks for the SPD manifold | 2026 |
| <ul style="list-style-type: none">• Authored and released an open-source Python package on the Python Package Index implementing the complete EGN architecture, including equivariant mapping, Riemannian pooling, geometric bias, and geodesic attention layers, with a stable matrix-logarithm backward pass.• Distributed under pip install egnlb; source at github.com/kraihan/EGN. | |

TEACHING EXPERIENCE

Lecturer	August 2024 – Present
<i>Department of Electrical and Electronic Engineering, North Western University, Khulna, Bangladesh</i>	

- Digital Image Processing (EEE-4255): deliver lectures on core theory, conduct laboratory sessions, design assignments, and supervise student projects in computer vision applications.
- Digital Signal Processing (EEE-4115): teach theoretical foundations and design assignments centered on biomedical signal analysis, including the discrete and fast Fourier transforms.
- Sessional on Digital Signal Processing (EEE-4116): conduct laboratory sessions and supervise student projects in EEG signal processing.
- Basic Electrical Circuits (EEE-1221): teach fundamental laws, network theorems, and circuit analysis, and develop the accompanying assignment sequence.
- Supervise three undergraduate thesis students from problem formulation through manuscript submission, teaching programming, classical machine learning, deep learning, and signal processing.

ACADEMIC SERVICE

Batch Advisor, third-year cohort, Department of EEE, North Western University	2025 – Present
Course Coordinator, Department of EEE, North Western University	2024 – Present
Co-founder, departmental research group, North Western University	2024 – Present
Vice President, KUET Mathematics Club	February 2023 – February 2024
Ambassador, International Youth Math Challenge – mathematics outreach in Bangladesh	2021

HONORS AND AWARDS

Honorable Mention, International Mathematics Competition, Bulgaria – 589 participants worldwide	2021
First participant from Bangladesh, 27th International Mathematics Competition	2020
Gold Medal, International Youth Math Challenge – top 1% of 15,000 competitors	2020, 2021
Excellence Award, IYMC Ambassador Program – fifth among 580 ambassadors	2021
Best Paper Award, IEEE Conference on Signal Processing, Information and Communication Systems	2024
Silver and Bronze Medals, International Astronomy and Astrophysics Competition	2022, 2021
Best Project Award, Electrical and Electronic Project Design Showcase, KUET	2023
University Dean's Award, KUET – second and fourth academic years	2021, 2024
University Merit Scholarship, KUET – awarded seven times for departmental ranking	2020 – 2024

TECHNICAL PROFICIENCIES AND LANGUAGES

Programming: Python, C++, MATLAB

Frameworks and libraries: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas, SciPy, OpenCV, MNE, EEGLAB, Matplotlib, Seaborn, Flask

Tools: Git and GitHub, LaTeX, Linux

Methods: generative modeling, flow matching and diffusion models, optimal transport, Riemannian and SPD-manifold learning, graph neural networks, low-level computer vision, EEG signal processing

Languages: Bengali (native), English (professional working proficiency)

REFERENCES

Dr. Mohiuddin Ahmad

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Relationship: undergraduate thesis advisor

Dr. Kalyan Kumar Halder

Professor, Department of Electrical and Electronic Engineering
Khulna University of Engineering and Technology, Khulna 9203, Bangladesh
Telephone: (+880) 1710-752236 · Email: kalyan@eee.kuet.ac.bd
Relationship: course instructor and mentor for International Mathematics Competition preparation